We will discuss these recipes in detail in the next few sections, while we also detail the implementation of these caches.

7.4 Getting Rid of Stale Data: LRU cache

The first type of cache, or rather of eviction policy, that we are going to describe is the LRU cache, purging each time the cache's **l**east **r**ecently **u**sed entry.

What do we need to implement this data structure? The operations we want to optimize for are:

- Storing an entry given the name of a company (obviously).
- Checking if there is an entry stored for a name.
- Retrieving an entry by name.
- Getting the number of elements stored.
- Getting the oldest entry and purging it from the cache.

As we have explained in section 7.3, this data structure will only be able to store a certain number of elements at the same time. The actual size of the cache can be decided on creation and, depending on the implementation, even be changed dynamically. We will denote with `N` the number of elements that can be stored on the cache. When the cache is already at its full capacity, if we need to add a new entry, we will have to remove one of the existing ones, and in particular in this case we will remove the one that was least recently used (hence the name).